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Barad Badri

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Mechanical Engineering, Specializing in Computing — *Sep. 2026 – Present*

A.Y. Jackson Secondary School — Toronto ON
Senior Editor of Yearbook, 2X Chemistry Award Winner, President of Engineering Club — *Sep 2022 – June 2026*

Experience

Architectural Assistant — Jan 2025 – June 2025
ArchiFad Architecture and Interior Design — *Aurora, ON*

- Created mechanical designs and floor plans using AutoCAD, SOLIDWORKS, FUSION360
- Researched materials, components, and tools required to prototype designs
- Communicated with clients to ensure expectations were met and the designs satisfied the criteria of customer expectation

Chef — Apr. 2026 – Sep 2026
Shoeless Joes — *Richmond Hill, ON*

- Worked alongside a brigade to create dishes and ensure high quality food is given to customers
- Preparing food, handling food, and storing food

Projects

Pneumatic Piston Engine Prototype | *LEGO, AutoCAD* — June 2025

- Created a prototype of a two-cylinder electric pneumatic piston engine using LEGO, scrap metal, and vacuum
- Used AutoCAD to design small components that would improve the engine
- Created a transmission system that would switch gears to control torque and speed

High School Yearbook | *Photoshop, Friesens CONNECTME* — Sep 2025 – Jun 2026

- Lead a team of 30 to create a yearbook that captures our school spirit
- Managed a $30,000 project budget, overseeing labour allocation, financial planning, and project execution alongside the Friesens company.
- Delegated tasks among the team to ensure deadlines were met

Pinball Machine | *AutoCAD, SOLIDWORKS, TinkerCAD* — Dec 2025 – Jan 2026

- Created a fully functioning pinball machine with poppers, electronic obstacles, and a functional point system
- Programmed an Arduino and connected it to several components (Solenoids, conductors, etc.,) to make the machine electronically functional
- Made the frame and various components using scrap wood, scrap metal, and 3D printing
- Made the pinball visually appealing with the use of dynamic LED's, where lights reacted to the balls movement using the Arduino

Technical Skills

Languages: Python, C/C++, HTML/CSS
CAD: AutoCAD, SOLIDWORKS, Fusion360, TinkerCAD
Design Tools: Photoshop, Illustrator, After Effects, Premier Pro
Other Skills: Bicycle Repair, Car Repair